

Every Tool in Its Place: Interaction and Collaboration with Robotic Drawers

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Abstract

In this study, we examined how participants (N=20) interacted and collaborated with a set of robotic drawers to accomplish a building task. The drawers' behavior varied along two variables – proactive/reactive and expressive/non-expressive motions. The results of our study indicated that participants considered an expressive robot to be more involved and interested in the interaction. They also found that while proactive or expressive robots could dominate the interaction, proactivity might negatively affect the participants' perception of their social status relative to that of the robot's, while expressiveness did not. This shows the importance of utilizing expressive movements when designing robots that collaborate with human users.

Introduction

Robotic systems, particularly domestic household robots, have become increasingly prevalent in our daily lives. As household robots continue to become more integrated into our homes and workspaces, we will need to interact and collaborate with them. However, Brazeal et al. (2004) indicated that both robots and human users had traditionally treated each other more like obstacles rather than social figures with whom to collaborate. Additionally, when people engage with robots to collaboratively accomplish a task, the means of interaction are often non-verbal and task-based. Thus, designing robots that can effectively work as a team with human users still remains an important challenge.

Our current research focuses on improving the engagement with interactive furniture and appliances, especially in the study of human gestures and other forms of non-verbal communication. In this study, we explored the interactions between human users and a set of robotic drawers that could collaborate together to accomplish a building task. We were interested in two variables that governed the robot's behavior. The first variable was the robot's expressiveness, with the robot being either



Figure 1: Robotic Drawers Performing an Animation.

non-expressive or expressive. Brazeal et al. (2003) noted the importance of a robot's expressivity in providing effective communication. In our study, we defined a non-expressive robot as one that performed only basic and fundamental movements to help complete a task. Conversely, an expressive robot carried out additional movements and animations. The second variable was the proactivity of the robot, with the robot being either reactive or proactive. Fink et al. (2014) and Mubin et al (2014) indicated that proactivity could be an important variable, especially when a robot worked together with human users. We defined a reactive robot to be one that responded and executed actions according to the user's gestures, while a proactive robot anticipated a user's impending needs and initiated an action to complete a task. To explore the effects of these two variables on interactions, we performed a controlled study.

Background

Related Work

There has been extensive research on different methods of improving the social interactions between robots and human users. One branch of research explores the use of anthropomorphism. Fink et al. (2012) incorporated several anthropomorphic elements into a robot to create more positive interactions. Osawa et al. (2007) examined the utilization of anthropomorphic features and frameworks for

domestic robots. The study found that the implementation of anthropomorphism could help robots facilitate better communication with human users. Though, one of the challenges with using anthropomorphism in task based applications is that the form factor of the robot is often dictated by the task. As designers may have limited ability to add any anthropomorphic features and forms to the robot, this then becomes problematic.

Another branch of research focuses more on gestures and different forms of nonverbal communication. Hoffman et al. (2013) indicated that most people were highly sensitive to physical movements. Hence, this might allow robots with well-designed motions to engage and communicate more effectively with human users. Sirkin et al. (2012) examined the use of both on screen and off screen motions for telepresence robotics. The study found that the additional motions displayed by the telepresence robots improved the understanding of messages and the sense of collaboration. Depending on the application, the ability to perform motions may already be present in the robot for pragmatic purposes. So, it may be possible to augment these robots for communicative and expressive purposes.

Prior Work

In our previous study (Mok et al. 2014), we examined how to create appropriate interactions for non-anthropomorphic robots, particularly with interactive furniture and appliances. Given a user's current actions and emotions, we were interested in how the robot should behave. So, we used video prototypes that explored the interactions between a human user and a set of robotic drawers. Through gestures and nonverbal communication, we had the user and the robot each display one of five possible emotional states – Angry, Happy, Indifferent, Sad, and Timid. These video prototypes were then utilized in a Qualtrics survey with participants recruited from Amazon Mechanical Turk, a crowdsourcing service. After watching the video prototypes, the participants (N=40) then answered questions pertaining to how they perceived the interaction between the user and the robot. Participants significantly preferred empathetic drawers that displayed emotions matching those of the user to the drawers that displayed emotions orthogonal to the user's emotions. Although neutral drawers were not as preferable as empathetic drawers, they were also significantly more preferable than drawers displaying orthogonal emotions. The results were reminiscent of Nass et al. (2004) in which emotional pairing of a car navigational system's voice to the driver's current emotional state significantly affected the performance of the driver. This reveals the importance of designing robots to display appropriate emotions.

Wizard of Oz Techniques

While it is ideal to have a fully functional autonomous robot in the field for our research, it will require a significant amount of time and effort to create this robust system. To examine how human users collaborate with a

set of robotic drawers, we need the AI of the system to always be able to interpret and understand the user's situation. This can be challenging when testing certain conditions. One example is the proactive robot which needs to sense that a certain step has been completed and that the user is ready to move on.

It was our goal to first evaluate how people in a household environment would interact with our robot. Since examining human robot interactions would be an iterative process that required the robot to be versatile and reliable, we proposed to use Wizard of Oz techniques to do our study. By utilizing a trained human teleoperator, we could still examine many of the questions we had without spending substantial amounts of time to develop an autonomous system. Prior work in this field, including Weiss et al. (2008) that explored public acceptance of robots, have demonstrated the usefulness of Wizard of Oz in exploring user experience factors. Similarly, Fink et al. (2014) used Wizard of Oz techniques to observe how a robot could interact with and encourage children to put away their toys.

Methodology

Physical Prototype

The robotic drawers prototype was fabricated from a standard Ikea MICKE 4-drawer unit. To allow the drawers to perform consistent and repeatable motions, the top 3 drawers were retrofitted with DC motors on a rack and pinion system. Additionally, spring-loaded rotational encoders, mounted against the drawers' frame allowed us to track each drawer's position. The bottom drawer was taped off from access during the study and contained parts/hardware that drove the system. The actuation was controlled by an Arduino microcontroller communicating with a local client program over a USB cable (See Fig.1).

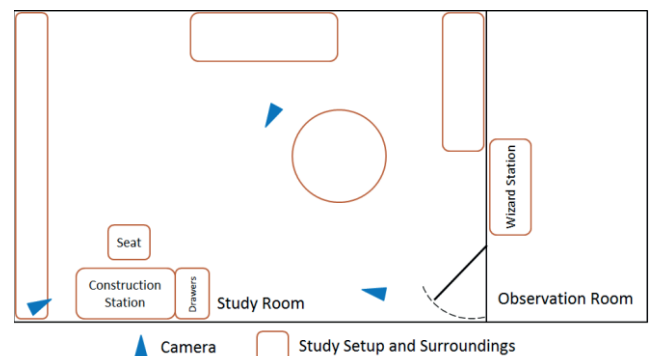


Figure 2: Diagram of System Setup

The local client program provided a Wizard of Oz style remote control for the drawers. A human "Wizard" in an adjacent room could remotely operate the drawers prototype via keyboard hotkey (Fig. 2). The UI presented the Wizard with 15 buttons and each button was bounded

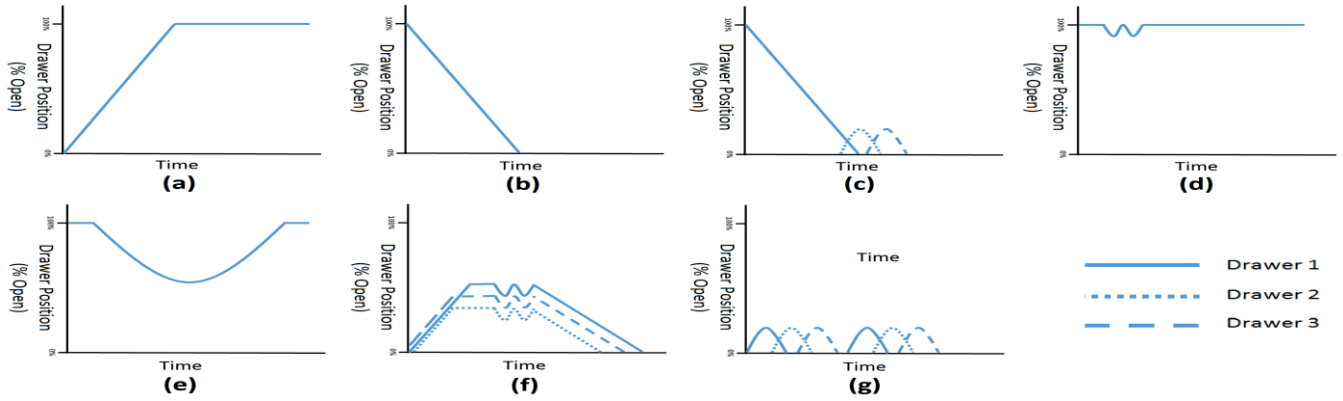


Figure 3: Animations of Drawers: (a) Simple Open (b) Simple Close (c) Flair Close (d) Wiggle (e) Beckon (f) Chuckle (g) Happy.

to a pre-programmed sequence of actuations, or animation, for the drawers to execute. The Wizard had a one-way audio/video feed of both the drawers and participants, which was needed to control the drawers appropriately.

Study Design

This research study explored how a robot's expressivity and proactivity affected human-robot turn-taking. To examine this, we devised a Wizard of Oz study involving a set of robotic drawers. We conducted a between-subjects test with four scenarios. Each scenario followed the same script, but varied two variables yielding a 2x2 factorial design. The first variable was the expressivity of the drawers. A non-expressive set of drawers would only open and close drawers whereas an expressive set of drawers would express sentiment with additional animations. The second variable was the proactivity of the drawers. In the proactive case, the drawers would lead by initiating interactions. While in the reactive case, the drawers would wait for gestures by the subject before reacting. Below we described the drawers prototype and the study protocol.

Animations

We wanted to have robotic drawers that appeared to be either non-expressive or expressive. Thus, animations were modeled accordingly to achieve this effect. The animations were designed through improvisation sessions with motion experts, similar to Sirkin et al. (2014). For a non-expressive drawer, we wanted to use the most basic and functional animations to interact with the participant. Therefore, there were only four animations. Three of these animations were used to perform a Simple Open (Fig. 3a) for each of the three drawers. For this animation, each individual drawer opened at a constant speed. The last animation was used to perform a Simple Close (Fig. 3b), with all the drawers closing at the same constant speed and remaining locked after closed.

For an expressive drawers, we added flair to the closing animation and more communicative gestures. The closing animation was now specific to the drawer that was currently opened. This would cause the other drawers to

open and close slightly at the end of the animation, mimicking a ripple effect. This is called the Close Flair (Fig. 3c) animation. We also added animations that suggested the drawers wanted something put back in or taken out of an opened drawer. To indicate the former, the drawers had a Wiggle (Fig. 3d) animation in which an opened drawer wiggled in and out slightly two times. To indicate the latter, there was a Beckon (Fig. 3e) animation in which an opened drawer closed half way and reopened at half speed. Lastly, we also added two animations to suggest sentiment, Chuckle and Happy. The Chuckle (Fig. 3f) animation was where all the drawers opened to random positions, wiggled twice, and closed. The Happy (Fig. 3g) animation involved all the drawers mimicking a ripple effect traveling down the drawers twice.

Participants

We recruited a total of 20 participants from the Stanford undergraduate and graduate student population and collected data over five days. Exactly 35% of participants were male and 65% were female. The ages of the participants ranged from 18 to 26 ($M=20.8$, $SD=3.03$). Each session required an average time of 45 minutes to complete and participants were compensated with a gift certificate.

Procedure

Participants were first asked to take a seat at the construction station (Fig. 2). They were then given an introduction to the study. The task for the participants was to assemble a cube made from six acrylic sheets and an 80/20 aluminum frame (Fig. 4). Each face of the cube required the participant to use a different type of fastener (Flathead, Hex Bolt, Philips, Spanner, Socket Cap, and Security Torx) to attach it to the frame. The participants were informed that the tools needed to complete this task were stored inside the drawers. Tools were distributed across the top three drawer and marked using white outlines so they could be returned to their original locations after the participant was done with them.

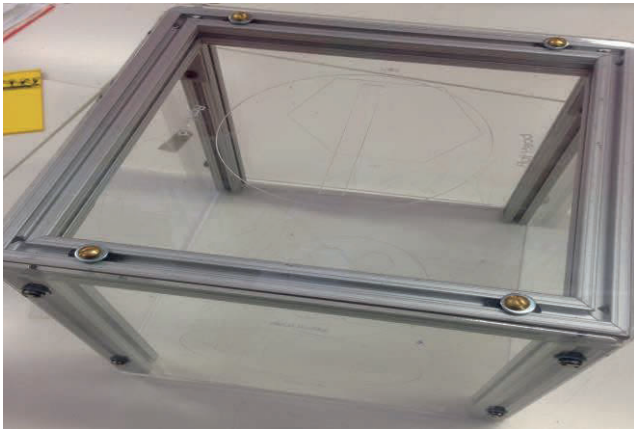


Figure 4: Acrylic and Aluminum Cube Assembled Using 6 Different Types of Fasteners and Tools In Different Drawers.

Depending on whether the participants were in the proactive or the reactive condition, they were told that “the drawers will provide you with the tools that you will need” or “you may gesture to the drawers to acquire the tools that you will need” respectively. Participants were also requested to take out a maximum of two tools at a time and build the cube by following the instructions in order. In earlier pilot testing, participants would often take out all of the tools at once, greatly changing the interactions with the drawers. Instilling these constraints allowed the interactions to be more structured and required the participants to interact with the drawers for at least a certain number of times. After all the instructions, the participants were allowed to then begin the task. When the task was finished, participants were given a questionnaire and an interview.

Wizard’s Responsibilities

Throughout the study, only one Wizard, who was trained to strictly follow the protocol, was used. The Wizard was tasked with actuating the set of drawers and performing the correct animations at the correct times. This was done by using the flow chart in Fig. 5. Given that the participant followed the order of tasks specified in the instructions and the “two tools” rule, the wizard would need to perform at least eight total tool exchanges (take and return). How these interactions specifically occurred would depend on the expressivity and proactivity conditions.

Along the first variable of expressivity, the wizard had to use different animations to convey the condition. If the drawers were non-expressive, the wizard would need to use the Simple Open animation when a tool needed to be taken from or returned to the drawer by the participant. After the tool was taken or returned, the wizard would then use the Simple Close animation. However, when a tool needed to be taken from or returned to an expressive drawer, the wizard would open a drawer using the Simple Open animation followed by the Wiggle animation. Similarly, the wizard would use the Close Flair animation instead of the Simple Close. In addition, the wizard would need to make the drawers to express certain sentiments. If

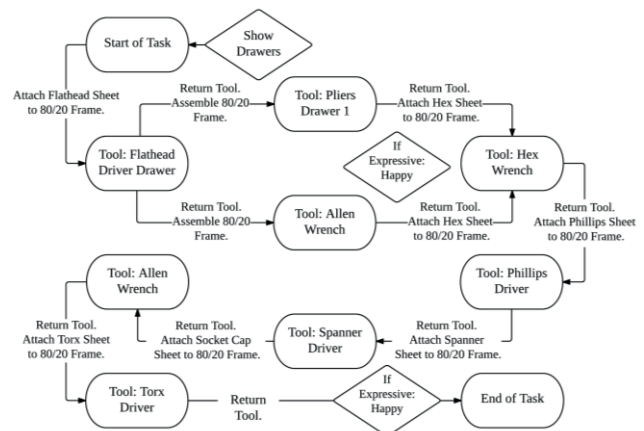


Figure 5: Flow Chart of Wizard's Task

the participant at any time made a mistake, such as dropping a fastener or picking up the wrong tool, the wizard would use the drawers’ Chuckle animation. Also, when the drawers congratulated the participant for completing milestones, the wizard would use the Happy animation. This was done when the frame was assembled and when the cube was completed.

On the second variable of proactivity, the wizard needed to change the behavior of the drawer based on the condition. If the drawers were proactive, the wizard would need to anticipate when the participant required a tool (based on where they were in the instructions) and opened that drawer with the tool in advance. The wizard would then close drawers when the participant took the tool. If the subject had two tools already and required a third one, the wizard would open one of the two former drawers to ask for a tool back. If the drawers were reactive, they only opened and closed based on the participant’s gestures. So, the subject needed to indicate to the drawers which one he/she wanted to open. The wizard would then open the drawer and keep it open until the participant gestured for it to be closed or after 10 seconds had passed. So, the subjects in the reactive scenario had more freedom as they were allowed to both request for a wrong tool and violate the “two tools” rule. The participants in the proactive scenario were coerced to follow the drawers’ lead, hence, they would not violate the above rule or request for a wrong tool.

Questionnaire

To help evaluate how the participants perceived their interactions with a set of drawers, we adapted and utilized the Dillard’s Relational Message Scale from Dillard et al. (1999). The relational factors that we incorporated were immediacy, affect, similarity, receptivity, composure, dominance, and equality. We adapted 26 statements regarding the participants’ interaction with the robot from the above relational factors. Participants were asked how much they agreed or disagreed with each of these statements based on a Seven-Point Likert Scale

(1=Strongly Disagree, 7=Strongly Agree). Comments were also collected.

Analysis and Results

First, the sets of Qualtrics responses were imported into *R* for statistical analysis. We used a two way ANOVA Test to determine if either the proactivity variable or the expressiveness variable had a significant effect on how the robot interactions were perceived by the participants. The questions that appear to show a strong or significant difference in response with respect to either proactivity or expressiveness are summarized in the following tables. Statistically significant results are highlighted in bold. Several other suggestive, though not statistically significant (to $p < 0.05$) results are also included. We will represent the conditions as follows: 1=Expressive and Reactive; 2=Expressive and Proactive; 3=Non-expressive and Reactive; 4=Non-expressive and Proactive. The Mean (M_x) and Standard Deviation (SD_x) of the Condition X will be presented in this format.

Immediacy and Affect

Questions from the Immediacy relational factor are related to the engagement and involvement of the interaction. Similarly, those of the Affect relational factor are related to the interest of the interaction.

Question	Proactivity	Expressiveness
The drawers found the interaction stimulating.	$F(1,16) = 1.39$ $p = 0.256$	$F(1,16) = 20.1$ $p < 0.01$
The drawers were intensely involved in the interaction.	$F(1,16) = 0.269$ $p = 0.611$	$F(1,16) = 3.61$ $p = 0.076$
The drawers were interested in interacting with me	$F(1,16) = 0.138$ $p = 0.715$	$F(1,16) = 3.45$ $p = 0.082$

Table 1: Immediacy and Affect Relational Factors

We can see that drawers exhibiting expressiveness were perceived by the participants to be more engaged and interested in the interaction compared to non-expressive drawers. Concerning whether it appeared that “The drawers found the interaction stimulating,” Expressive drawers ($M_1=6$, $SD_1=0.71$; $M_2=5.4$, $SD_2=0.89$) have greater Immediacy than Non-expressive drawers ($M_3=4$, $SD_3=1.2$; $M_4=3.6$, $SD_4=0.89$). When participants were asked if “The drawers were intensely involved,” similar results are found with Expressive drawers ($M_1=5.8$, $SD_1=0.84$; $M_2=5.8$, $SD_2=1.8$) having greater Immediacy than Non-expressive drawers ($M_3=4.4$, $SD_3=1.5$; $M_4=5$, $SD_4=0.71$). Finally, regarding if “The drawers were interested in interacting with me,” the Expressive drawers ($M_1=5.4$, $SD_1=1.1$; $M_2=5$, $SD_2=1.4$) similarly have greater Affect than Non-expressive drawers ($M_3=3$, $SD_3=1.3$; $M_4=3.4$, $SD_4=0.89$). When looking closer at the mean response for each condition for this question, we see an

interesting interaction effect. It seems that a set of reactive drawers appeared particularly to be more interested and involved in the interaction when they were also expressive. However overall, being proactive or reactive does not appear to make a significant difference on the perception of Immediacy and Affect.

Similarity

Questions from the Similarity relational factor are related to the familiarity and friendliness of the interaction.

Question	Proactivity	Expressiveness
The drawers seemed to desire further interaction	$F(1,16) = 0.023$ $p = 0.882$	$F(1,16) = 14.2$ $p < 0.01$
The drawers acted like we were good friends	$F(1,16) = 4.41$ $p = 0.053$	$F(1,16) = 4.31$ $p = 0.055$
The drawers tried to move the interaction to a deeper level.	$F(1,16) = 0.191$ $p = 0.668$	$F(1,16) = 2.57$ $p = 0.128$

Table 2: Similarity Relational Factor

Again it appears that the expressive drawers were perceived to be more familiar and friendlier than non-expressive drawers. Looking at the mean response, we see another interesting interaction effect. When reviewing the question “The drawers acted like good friends”, drawers with the combination of expressive and reactive ($M_1=5.4$, $SD_1=0.55$) create an effect that is very different from that of other drawers ($M_2=3.2$, $SD_2=1.3$; $M_3=3$, $SD_3=1.4$; $M_4=3.4$, $SD_4=0.89$). Other than this, proactivity again appears to not produce any significant difference.

Domination

The Domination relational measure and its questions help determine the leader or influential figure in the interaction.

Question	Proactivity	Expressiveness
The drawers tried to dominate me.	$F(1,16) = 3.35$ $p = 0.086$	$F(1,16) = 6.84$ $p = 0.019$
The drawers tried to persuade me.	$F(1,16) = 16.4$ $p < 0.01$	$F(1,16) = 4.88$ $p = 0.042$
The drawers tried to control the interaction.	$F(1,16) = 18.4$ $p < 0.01$	$F(1,16) = 2.38$ $p = 0.142$
The drawers did not attempt to influence me.	$F(1,16) = 18.8$ $p < 0.01$	$F(1,16) = 26.6$ $p < 0.01$

Table 3: Domination Relational Factor

We can see that either exhibiting expressive or proactive can lead to the drawers appearing more dominant and

influential. This is expected for proactive since the drawers were always in a leading role and would not yield to the participants' gestures. For expressive drawers, participants might feel that the animations were trying to get their attention, thus influencing and persuading them. Also note that the question of "The drawers did not attempt to influence me" is reverse coded.

Equality

The Equality relational measure concerns whether both parties in the interaction feel that they are being treated on an equivalent social standing/status. So the questions can help determine if any inequality had occurred.

Question	Proactivity	Expressiveness
The drawers did not treat me as an equal.	$F(1,16) = 2.33$ $p = 0.147$	$F(1,16) = 0.582$ $p = 0.457$
The drawers considered us equals	$F(1,16) = 7.94$ $p = 0.013$	$F(1,16) = 0.025$ $p = 0.878$

Table 4: Equality Relational Factor

Despite the results from the dominance relation factor, we can see that expressiveness does not have an effect, with no strong difference in equality. This is interesting, given the results of the Dominance relational factor. Conversely, proactivity has a very notable difference. Regarding "The drawers considered us equal," Proactive drawers ($M_2=3.8$, $SD_2=1.1$; $M_4=2.8$, $SD_4=0.96$) created a greater feeling of inequality compared to Reactive drawers ($M_1=4.2$, $SD_1=0.84$; $M_3=5$, $SD_3=0.96$). Examining the mean responses, participants felt that the proactive drawers did not treat them as equals, while reactive drawer did. Also, note that the responses to the question of "The drawers did not treat me as an equal" is reverse coded for analysis. So a similar result, proactivity caused participants to have a greater feelings of inequality, is found.

Discussion

One of the findings is the ability for the robot to appear to be persuasive and influential to the participant when it is either expressive or proactive. We see strong differences with regards to questions about the Dominance relational factor. However, when the robot was proactive in the interaction, it caused participants to feel that the robot did not treat them as equals, as indicated by the strong difference in the Equality relation factor. According to one of the post study interviews, the robot appeared very much to be "like a boss" figure, relegating people to a lower social standing. In one case, a participant indicated that "I was the builder, the drawers should not command me to do things. I will do it when I am ready." Another participant felt that "It was distracting when I was trying to understand what's going on, I knew it was trying to get me to get the tool for the next step, but I didn't know what to do yet. So I just ignored it until I was ready". Conversely, when the robot exhibited expressiveness, it did not create this feeling

of inequality. One participant noted that "It was like a fiery little Scottish Terrier trying to pull me its way." Several other participants similarly noted that it was "like a pet". So, by incorporating an expressive nature into the robot's actions, the robot can still help lead interactions while making the participant feel like an equal. It is important as this difference in status can cause frustration and discomfort for the user of the robot.

Through post study interviews, we also found that the participants were able to readily determine the intents and messages of the robot's actions. As noted in some of the quotes above, participants knew that the robot was trying to get them to move onto the next step by either asking them to take or return a tool. However, as also noted, the participants wanted to work on the task at their own pace. So, they often ignored the robot's suggestions until they were ready to move onto the next step. Another important insight that we observed is the effectiveness of the expressive movement displays, despite what the participant was focused on. Reviewing the video data, participants were often focused elsewhere, often on the task, when an expressive drawer performed animations. For example, participants were often looking at the cube when a Happy animation was occurring. However, in the post study interview, participants consistently recalled seeing the animation, with a large percentage realizing that "it was trying to congratulate me." Similar observations were seen with the Chuckle animation. One participant noticed that "it was warning me of an error." Even without directly looking at the expressive drawers, participants still experienced the drawers' movements. This finding reaffirms Hoffman et al. (2014) in that humans are sensitive to movement and that small well designed motion can be used to communicate.

Additionally, we found another insight from monitoring the participants interacting with the reactive drawers. As they were not prompted with how they should interact with the drawers, the subjects used a variety of different physical gestures to communicate with the drawers. These included touching, waving, and performing swiping motions. Some participants even tried talking to the drawer in addition to gesturing. It is also interesting that participants quickly created a mental model of how to interact with the drawer and would not try different gestures or motions, even in rare cases when the drawers did not quickly react.

Conclusion

This Wizard of Oz controlled study shows that having a robot perform expressive movements greatly affect how an interaction is perceived by the user. Participants find that expressiveness allows the robot to seem more engaged, involved, and interested in the interaction compared to a more functional non-expressive robot. Similarly, the expressive robot also appears friendlier to the participants. Proactivity seems to have no effect on how engaged the robot appears, but a reactive and expressive robot does

appear to the participant to be like a good friend. We also note that a proactive robot seems to be more dominant than reactive robots, to the point where the participant feels a sense of inequality. However we find that expressiveness can help temper this feeling, as an expressive robot can seem dominating but still of a similar social standing. These findings can help create design guidelines for robots that will collaborate with human users.

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Reference

- Breazeal, C. (2003). Emotion and sociable humanoid robots. *International Journal of Human-Computer Studies*, 59(1), 119-155.
- Breazeal, C. (2004). Social interactions in HRI: the robot view. *Systems, Man, and Cybernetics, Part C: Applications and Reviews*, IEEE Transactions on, 34(2), 181-186.
- Dillard, J. P., Solomon, D. H., & Palmer, M. T. (1999) Structuring the concept of relational communication. *Communications Monographs*, 66(1), 49-65.
- Fink, J. (2012). Anthropomorphism and human likeness in the design of robots and human-robot interaction. In *Social Robotics* (pp. 199-208). Springer Berlin Heidelberg.
- Fink, J., Lemaignan, S., Dillenbourg, P., Rétornaz, P., Vaussard, F., Berthoud, A., ... & Franinović, K. (2014, March). Which robot behavior can motivate children to tidy up their toys?: design and evaluation of ranger. In *Proceedings of the 2014 ACM/IEEE international conference on Human-robot interaction* (pp. 439-446). ACM.
- Hoffman, G., & Ju, W. (2014). Designing Robots With Movement in Mind. *Journal of Human-Robot Interaction*, 3(1), 89-122.
- Mok, B. K., Yang, S., Sirkin, D., & Ju, W. (2014, March). Empathy: interactions with emotive robotic drawers. In *Proceedings of the 2014 ACM/IEEE international conference on Human-robot interaction*. ACM.
- Mubin, O., D'Arcy, T., Murtaza, G., Simoff, S., Stanton, C., & Stevens, C. (2014, August). Active or passive?: Investigating the impact of robot role in meetings. In *Robot and Human Interactive Communication, 2014 RO-MAN: The 23rd IEEE International Symposium on* (pp. 580-585). IEEE.
- Nass, C., Jonsson, I. M., Harris, H., Reaves, B., Endo, J., Brave, S., & Takayama, L. (2005, April). Improving automotive safety by pairing driver emotion and car voice emotion. In *CHI'05 Extended Abstracts on Human Factors in Computing Systems* (pp. 1973-1976). ACM.
- Osawa, H., Mukai, J., & Imai, M. (2007). Anthropomorphization Framework for Human-Object Communication. *JACIII*, 11(8), 1007-1014.
- Sirkin, D., & Ju, W. (2012, March). Consistency in physical and on-screen action improves perceptions of telepresence robots. In *Proceedings of the seventh annual ACM/IEEE international conference on Human-Robot Interaction* (pp. 57-64). ACM.
- Sirkin, D., & Ju, W. (2014, October). Using Embodied Design Improvisation As A Design Research Tool. *International Conference on Human Behavior in Design*.
- Weiss, A., Bernhaupt, R., Tscheligi, M., Wollherr, D., Kuhnlenz, K., & Buss, M. (2008, August). A methodological variation for acceptance evaluation of human-robot interaction in public places. In *Robot and Human Interactive Communication, 2008. RO-MAN 2008. The 17th IEEE International Symposium on* (pp. 713-718). IEEE.